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Catalytic upgrading of ethanol to n-butanol: A progress in catalyst development

Xianyuan Wu, Geqian Fang, Yuqin Tong, Dahao Jiang*, Zhe Liang, Wenhua Leng, Liu Liu, Pengxiang Tu, Hongjing Wang, Jun Ni* and Xiaonian Li

Abstract: In view that n-butanol as fuel additive has more advantageous physico-chemical properties compared to ethanol, ethanol valorization to n-butanol via homogeneous or heterogeneous catalysis has received great attention over the last decades both in the scientific and industrial fields. This paper reviewed the recent progress in catalyst development for upgrading ethanol to n-butanol, which involves homogeneous catalysts such as Iridium and Ruthenium complexes catalysts and heterogeneous catalysts mainly including metal oxides, hydroxyapatite (HAP) and especially supported metal catalysts. The structure-performance relationship of catalysts and underlying reaction mechanism were critically examined. Future research directions on the catalyst design and improvement were also proposed.

1. Introduction

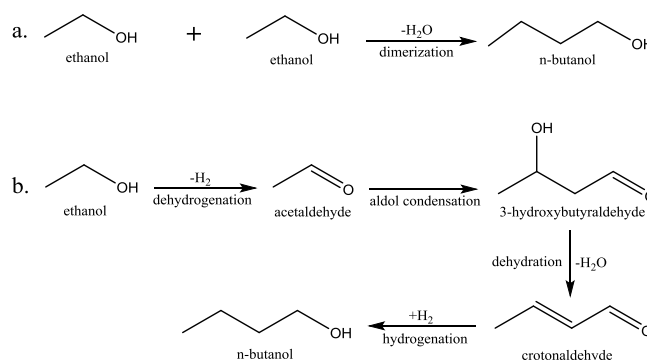
The interest in biofuels has increased considerably over the last decades, particularly in view of concerns over diminishing crude oil reserves and greenhouse gas emissions. Bio-ethanol, as a sustainable and clean biofuel, has been currently used as an additive to gasoline in the United States, Brazil, China, and some European countries.^[1] However, ethanol has some obvious problems as a fuel, including its high water solubility, low energy density and corrosion to engine structure. These disadvantages are generally mitigated by replacing ethanol with n-butanol, which has more superior fuel properties, such as higher energy density (~90% of that of gasoline), lower water adsorption, higher air-to-fuel ratio, and lower heat of vaporization^[2] (shown in Table 1), therefore is a highly promising biofuel in future.^[3] In addition to being an important biofuel, n-butanol and higher alcohols are also valuable chemical feedstocks and components for solvent, chemical intermediate and polymer material.^[3] Unfortunately, n-butanol is currently manufactured from fossil-based materials such as propylene and CO through the energy-intensive hydroformylation process (also known as the oxo process)^[4] by using a homogeneous cobalt or rhodium catalysts in industry. Additionally, although bio-butanol can be also obtained via a biological synthetic route using a *Clostridium* anaerobe to ferment starch and cellulosic materials, the overall n-butanol yield is less than

Table 1. The comparison of physico-chemical properties of gasoline, diesel, ethanol and n-butanol.

Fuel	Energy density (MJ/L)	Water solubility (%)	Air-to-fuel ratio	Heat of vaporization (MJ/Kg)	Octane number
Gasoline	32	-	14.6	0.36	91-99
Diesel	35.5	-	14.7	-	-
Ethanol	19.6	100	8.94	0.92	129
n-butanol	29.2	7	11.12	0.43	96

2%. An increasing cost of fermentation substrates, separation of the products, time consumption during disinfection of fermentation device, and especially high toxicity of solvents to *Clostridium* stains also largely limit the large-scale industrial production of bio-butanol.^[5]

On the other hand, the production of bio-ethanol from cellulosic residues will generate a large supply of bio-ethanol all over the world in the next decade.^[1] Moreover, n-butanol can be achieved by direct dimerization of two molecular ethanol (Scheme 1a) or by a multi-step tandem synthetic route from ethanol known as the Guerbet reaction (Scheme 1b) including a dehydrogenation of primary ethanol to form acetaldehyde, followed by aldol condensation before the resultant product is dehydrated and subsequently hydrogenated to produce n-butanol.^[6] The formed butanol can then react with itself or with ethanol to form other higher alcohols such as 1-hexanol, 2-ethyl-1-butanol and 1-octanol.



Scheme 1. (a) direct and (b) indirect approaches for n-butanol formation from ethanol.

[a] X. Wu,* G. Fang,* Y. Tong, D. Jiang, Z. Liang, W. Leng, L. Liu, P. Tu, H. Wang, J. Ni, X. Li.

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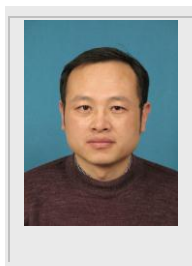
[*] These authors contributed equally to this work.

MINIREVIEW

Also, other by-products such as ether acetate, 1,1-diethoxyethane, diethyl ether and ethylene probably form by condensation, dehydration or dehydrogenation reaction of ethanol and/or acetaldehyde depending on the nature of the catalysts used.^[7] This direct or indirect butanol synthesis pathway from ethanol has been considered an economical and sustainable route. Thus, the upgrading of ethanol to n-butanol and other higher alcohols via homogeneous or heterogeneous catalytic conversion has currently received great attention both in the scientific and industrial fields regarding energy and chemicals sustainability.^[8]

The present review aims to provide a summary on the recent progress in catalyst development for upgrading ethanol to n-butanol with the emphasis on the structure-performance relationship of catalysts and underlying reaction mechanism. A brief discussion of future research on catalyst design and improvement for upgrading ethanol to n-butanol is also provided.

Associate Prof. Dahao Jiang is a principle researcher at the Institute of Industrial Catalysis, Zhejiang University of Technology, China. His research group has been engaged in the development of nanostructured functional materials for catalytic conversion of biomass raw materials and biomass-derived platform compounds. His current research interest focuses on design and synthesis, characterization, and application of highly efficient heterogeneous catalysts for bio-ethanol conversion into n-butanol.



Associate Prof. Jun Ni obtained his Ph.D. in Chemical and Chemical Engineering from Queen's University of Belfast, UK. After that, he worked as Post-doctor for French National Center for Scientific Research (CNRS) in Laboratory for Catalysis & Spectrochemistry in 2009 and Research Fellow at the National University of Singapore during 2010-2012. He is currently employed at Zhejiang University of Technology, China. In the past decade, his research has focused on the development of heterogeneous catalysis for a wealth of application in various industries and in many domains linked to the biomass conversion and utilization.



2. Homogeneous Iridium and Ruthenium complexes catalysts

The transition from a petroleum-based raw materials to renewable resources for n-butanol synthesis is one of the key research challenges of the coming years. Homogeneous catalysis with high selectivity and activity has emerged as an important tool for this conversion process.^[8] The main type of homogeneous catalysts for upgrading ethanol to n-butanol and their catalytic performance

are presented in Table 2. In 2009, Ishii and co-workers reported an iridium complexes catalyst, [Ir(acac)(COD)] (acac: acetylacetonate, COD: 1,5-cyclooctadiene) could promote ethanol upgrading with 41% ethanol conversion and 51% selectivity to n-butanol in the presence of a phosphine ligand, the additive 1,7-octadiene and EtONa.^[9] However, this reaction was carried out in pure alcohol to avoid the detrimental effect of water.^[10] Subsequent report showed a [Ir(OAc)₃] catalyst with a ligand (bathophenanthroline disulfonic acid disodium salt) exhibited 26% selectivity to n-butanol at 52% ethanol conversion in water, but iridium precipitates could be observed.^[11] The ethanol upgrading to n-butanol in this homogeneous reaction system is proposed to proceed through the Guerbet reaction (shown in Scheme 1b): the precious metal (Ir) was responsible for the alcohol dehydrogenation and subsequent unsaturated aldehyde hydrogenation to higher alcohol, while the strong inorganic base (e.g. NaOH, EtONa) was responsible for the aldol coupling step. The resultant product could undergo further aldol condensation and hydrogenation to give longer chain alcohols, such as n-hexanol and n-octanol. However, for the industrial production of n-butanol, the catalytic activity, selectivity and stability towards n-butanol over iridium complexes catalysts should be greatly improved.

In 2013, Wass et al. reported a new family of ruthenium-based homogeneous catalysts which could offer a step forward in catalytic performance.^[12] The optimal ruthenium catalyst incorporating a bis(diphenylphosphanyl)methane ligand promoted the upgrading of ethanol to n-butanol with over 94% selectivity to n-butanol at 22% ethanol conversion. This remarkable feature of extremely high n-butanol selectivity over the ruthenium catalyst compared to that of previously reported iridium complexes catalysts is mainly ascribed to the effective control over the base-catalyzed acetaldehyde aldol reaction, so that only exclusive C₄ products were obtained rather than higher carbon products produced by further aldol coupling via the same Guerbet process. However, the catalytic activity decreased with time on stream, because these ruthenium complexes catalysts decomposed to nanoparticulate metal over the course of the reaction. In this case, the decrease in catalytic activity is probably attributed to the weak water tolerance of ligands which decomposed in the process of reaction. New water tolerant catalysts are thus necessary considering the Guerbet reaction occurs in the presence of water. A recent report by the same group showed several ruthenium complexes catalysts with mixed donor phosphine-amine (P-N) ligands for the conversion of ethanol to n-butanol exhibited 92.7% n-butanol selectivity and up to 31.4% ethanol conversion with turnover numbers (TON) beyond 300.^[13] This kind of mixed donor P-N ligands not only had a superiority in being deprotonated in hydrogen transfer reactions but also enhanced the water tolerance of ruthenium catalysts relative to that of previously reported diphosphine systems.^[12] Although excellent selectivity of n-butanol and higher conversion of ethanol were achieved, the mixed donor P-N ligands still suffered from decomposition in the process of reaction. The reason for the ligand decomposition is unknown, and more research work needs to be carried out.

MINIREVIEW

Table 2. Comparison of catalytic performance of homogeneous catalysts for upgrading ethanol to n-butanol reported in documentary.

Catalyst ^[a]	T (K)	Reaction time (h)	Ethanol Conversion (%)	Butanol Selectivity (%)	Butanol Yield (%)	TON	Ref.
[Ir(acac)(COD)], NaOEt	393	15	41	51	21	1220	[9]
[Ru-1], NaOEt	423	4	22.1	93.6	20.1	221	[12]
[Ru-2], NaOEt	423	4	31.4	92.7	28.1	314	[13]
[Ru-3], NaOEt	423	2	53	78	37	530	[14]
[Ru-4], NaOEt	383	168	18.2	86.1	14.6	18209	[16]
[Ir], [Tp [*] Ni(μ-OH)] ₂	423	24	37	99	34	185	[17]

[a] acac: acetylacetonate; COD: 1,5-cyclooctadiene; [Ru-1]: [RuCl(η⁶-p-cymene)(bis(diphenylphosphanyl)-methane)]Cl; [Ru-2]: [RuCl₂(η⁶-p-cymene)]₂, additive: 2-(diphenylphosphino)ethylamine; [Ru-3]: bis(pyridylimino)-isoidolate(bpi) ruthenium complexes, additive: PPh₃; [Ru-4]: acridine-based pincer complex [RuHCl(A-iPr-PNP)(CO)], iPr: 1,3-bis(2,6-diisopropylphenyl)-1,3-dihydro-2H-imidazol-2-ylidene, PNP: 2,6-bis(di-tert-butylphosphinomethyl)pyridine; [Ir]: Cp^{*}Ir[(2-OH-6-phenyl)pyridine]Cl; [Tp^{*}Ni(μ-OH)]₂, Tp^{*}: tris(3,5-dimethyl-pyrazolyl)borate).

More recently, Szymczak et al. reported an amide-derived N,N,N-Ru(II) complex capable of upgrading ethanol to n-butanol with the TON of 530.^[14] The ethanol conversion is as high as 53%, however, at the sacrifice of selectivity to n-butanol (~80%) in comparison to that of previously reported Ru complexes catalysts achieving >90% butanol selectivity.^[12,13] This improved catalytic activity is mainly ascribed to higher catalytic acceptorless alcohol dehydrogenation(AAD) and hydrogenation activity of the new Ru complexes catalyst. Their prior studies have already demonstrated this analogous Ru complexes catalyst capable of mediating reversible transformation between ketones and alcohols via sequential hydrogenation-acceptorless alcohol dehydrogenation.^[15] In addition to high catalytic activity, this ethanol upgrading to n-butanol reaction was carried out under ambient conditions with no loss in activity, indicating the Ru complexes catalyst is neither decomposed nor oxidized. To further improve the conversion of ethanol, Milstein et al. applied an acridine-based ruthenium pincer complexes, and up to 73.4% ethanol conversion and 60.3% selectivity towards n-butanol were achieved with 0.02 mol% catalyst. With the catalyst loading of 0.001 mol%, a record TON of 18209 could be reached.^[16] The mechanistic insight of this highly efficient catalyst and further improvements are underway.

Notably, Jones and co-workers reported a bifunctional iridium catalyst coupled with bulky nickel or copper hydroxides for the upgrading of ethanol to n-butanol via a tandem catalytic Guerbet process. This process produced n-butanol with a remarkably higher selectivity (>99%) at 37% ethanol conversion than that ever reported in the homogeneous reaction system.^[17] In this Guerbet reaction chains, the active centers of dehydrogenation and hydrogenation were mainly provided by a Cp^{*}Ir[(2-OH-6-phenyl)pyridine]Cl (Ir) catalyst which is an excellent homogeneous catalyst for the acceptorless dehydrogenation of primary alcohols and hydrogenation of α,β-unsaturated aldehydes.^[18] The critical step of aldol condensation was successfully performed by strong basic transition-metal hydroxide

complexes containing sterically encumbered ligands such as (Ni₂): [Tp^{*} Ni(μ-OH)]₂ or (Cu): (IPr)CuOH. Both of these two ligands showed excellent aldol coupling activity and gave complete C4 product selectivity. However, the inherent problems of non-reusability and the necessity of special handling precautions for air-sensitive metal complexes have restricted the utility of these procedures.

3. Heterogeneous catalysts

3.1. Metal oxide and zeolite catalysts

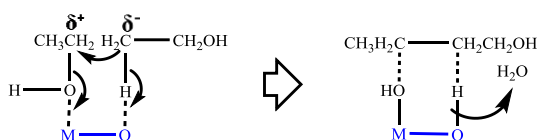
In view of the drawbacks of homogeneous catalysts such as discontinuity of production and potential environmental pollution concerning the use of soluble bases and organic solvents, heterogeneous catalysts and processes are extensively explored for the conversion of ethanol to n-butanol in recent years. Basic metal oxide catalysts have been widely applied in the ethanol upgrading reaction,^[6a,19] among which MgO exhibited the highest reaction activity (~56% ethanol conversion) and ~33% selectivity to n-butanol (723 K, 1 bar).^[6a] Attempts were made to enhance the selectivity of MgO catalyst by doping various basic metals (e.g. Ca, Ba) and transition metals (e.g. Zn, Ce, Zr, Pb, Sn). However, no improvement of catalytic performance was observed. In contrast, n-butanol selectivities were significantly decreased over these basic and transition metals modified MgO catalysts. This result is mainly attributed to the difference between the quantity of strong basic sites on MgO and that on modified MgO catalysts (characterized by TPD measures). The introduction of these doping metals largely decreased the quantity of strong basic sites required for abstraction of a β-H atom in ethanol. Thus, the strong basic sites are active for the ethanol condensation at high temperature. In addition, some gaseous light products were formed by thermal decomposition of ethanol at high temperature.

MINIREVIEW

Table 3. The influence of different densities of basic and Lewis acid sites on catalytic activities and selectivities to n-butanol and C₄ products.^[a]

Catalyst	Density of basic sites ($\mu\text{mol/g}$)			Density of Lewis acid sites ($\mu\text{mol/g}$)	Ethanol conversion (%)	Selectivity (%)		
	W	M	S			n-Butanol	C ₄ products	Acetaldehyde
Mg ₃ Al ₂ O ₇	21	27	-	0.2	40.5	22	68.8	6.8
Mg ₂ Al ₂ O ₇	13	18	-	0.2	33.2	34.2	55.6	22.2
MgO	19.4	15	29	-	5.2	-	-	69.8
Al ₂ O ₃	43.3	-	-	0.3	90.1	-	-	-

[a] Reaction conditions: 0.3 g catalyst, 623 K, atmospheric pressure, 12% ethanol/N₂ (vol. %) at 40 ml/min.

**Scheme 2.** The dimerization mechanism of two ethanol molecules.

Based on the distribution of various intermediates and the related works done early by Iglesia and co-workers,^[20] the ethanol upgrading to n-butanol on basic metal oxides at relatively high temperature is not proposed to proceed primarily through the Guerbet reaction (shown in Scheme 1b). Instead, this process is proceeded through a dimerization mechanism as seen in Scheme 2, in which a C_β-H bond in ethanol is activated by the basic metal oxide and the activated ethanol molecule subsequently condenses with another molecule of ethanol by dehydration to form n-butanol.

In addition to basic metal oxide catalysts, catalysis for ethanol conversion to n-butanol over catalysts containing both acid and basic sites is another area of interest. Yang and Meng^[21] conducted ethanol upgrading to n-butanol over basic monometallic and bimetallic modified 13X zeolite catalysts under reaction conditions: 693 K, 1 bar, W/F=5.6 g·h·mol⁻¹. Almost no n-butanol but 100% selectivity to gaseous light products were detected over monometallic modified zeolites, whereas the bimetallic modified Rb-LiX-zeolite catalyst showed better catalytic performance (ethanol conversion rate of $\sim 6 \times 10^{-4}$ mol·min⁻¹·g⁻¹ and $\sim 40\%$ n-butanol selectivity). The better catalytic activity of bimetallic modified zeolites than monometallic modified zeolites is attributed to a stronger basicity required for the activation of C_β-H bond in ethanol over bimetallic modified zeolite catalysts. Additionally, the rate of ethanol conversion linearly increased from 673 to 753 K, while the rate of n-butanol formation increased rapidly from 673 to 693 K and leveled off from 693-753 K, but followed by a decline at temperatures higher than 753 K. It suggests a high reaction temperature is detrimental to the formation of n-butanol. In this work, the ethanol coupling at high temperatures (>673 K) is also thought to proceed via dimerization mechanism of two ethanol molecules, which is similar to that of reported basic metal oxide catalysts (shown in Scheme 2).^[19,20]

Kozłowski and co-workers^[22] carried out ethanol conversion with varying loadings of Na modified ZrO₂ catalysts (613-673 K, 1.36-1.7 bar, $0.39\text{-}2.1 \mu\text{mol}_{\text{ethanol}}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$). The density of basic sites of catalysts increased with the addition of Na content (0.1-1 wt%), whereas the acid site density decreased at the same time. This decrease in acid site density markedly decreased the rate of dehydration of ethanol, as a result of which the overall n-butanol selectivity was improved with the addition of Na. In this reaction system crotonaldehyde as an important intermediate was observed, suggesting the Guerbet pathway is the predominant route of ethanol coupling to n-butanol (Scheme 1b). Despite the promotional effect of Na on the n-butanol selectivity, the rate of crotonaldehyde production derived from aldol condensation of acetaldehyde decreased with the increase in Na content. Possibly the introduction of Na led to a higher surface coverage of adsorbed intermediates such as ethoxide or hydroxide, which might inhibit the C-C coupling reaction of acetaldehyde. They also examined the effect of reaction temperature and found when the reaction temperature was raised from 613 to 673 K, the rate increment of acetaldehyde production was an order of magnitude higher than that of n-butanol production. This result clearly indicates the increase in reaction temperature is more preferable for the production of acetaldehyde compared to that of n-butanol.

Although the acidity of ZrO₂ and zeolite can be reduced by the incorporation of basic metal modifiers, several by-products such as ethylene and diethyl ether via catalytic dehydration of ethanol were still detected. Thus, it seems that both ZrO₂ and zeolite are not ideal candidates for ethanol upgrading to n-butanol at the moment. More investigations related to detailed reaction mechanism and the approach to improve n-butanol yield over ZrO₂- and zeolite-based catalysts should be further conducted.

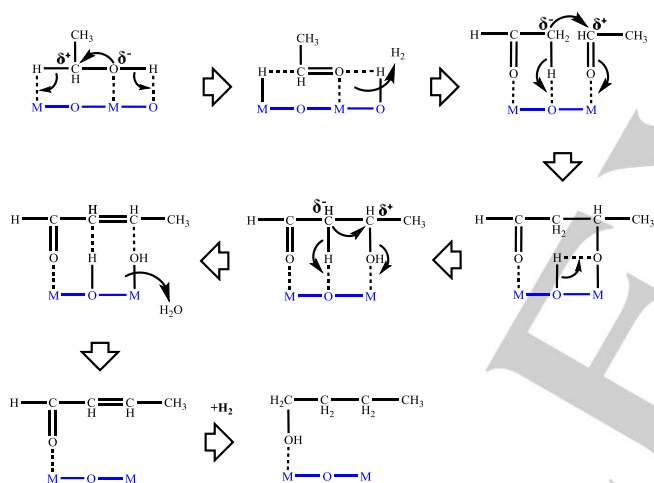
Mg-Al mixed oxide catalysts derived from hydrotalcite precursors (LDH) have received great attention as very promising catalysts for the catalytic ethanol upgrading process due to their high surface area, tunable acid-base properties and structural stability.^[23] Carvalho et al. employed Mg-Al mixed oxide catalysts derived from hydrotalcites including pairs of adjacent acid and basic sites such as Mⁿ⁺-O²⁻ (M=Mg, Al) for the conversion of ethanol to n-butanol.^[24] The reaction results are shown in Table 3 and detailed mechanism is shown in Scheme 3.^[25] As shown in Table 3, the catalyst with the Mg/Al ratio of 3 showed superior catalytic performance over previously reported mixed oxide catalysts. The reaction mechanism involves adjacent Lewis acid-basic site pairs playing a vital role in enhancing catalytic activity and C₄ products selectivity (Scheme 3). Specifically, one ethanol

MINIREVIEW

Table 4. Ethanol conversion and selectivity to n-butanol and other by-products over various M-Mg-Al-O mixed oxide catalysts (M=Cu, Pd, Ag, Mn, Fe, Sm, Yb).^[a]

Catalyst	Total number of basic/acid sites (mmol/g)	Ethanol conversion (%)	Selectivity (%)			Butanol yield (%)
			n-Butanol	Ethyl acetate	1,1-Diethoxyethane	
Cu ₁ MgAl ₃ O	1.85/1.94	2.5	43	2	35.4	1.08
Cu ₅ MgAl ₃ O	1.58/2.30	4.1	40	7.2	36.2	1.64
Cu ₁₀ MgAl ₃ O	1.20/1.46	4.5	28	12.5	43.4	1.26
Cu ₂₀ MgAl ₃ O	0.89/2.27	3.8	18	45.4	28.2	0.68
Pd ₅ MgAlO	1.94/1.74	3.8	72.7	4	2.5	2.76
Ag ₅ MgAlO	1.62/2.34	1.6	38.8	25.9	7.5	0.62
Mn ₅ MgAlO	1.36/3.37	0.7	53.3	2.9	6.9	0.37
Fe ₅ MgAlO	1.60/3.51	0.3	39.2	2.7	26.7	0.12
Sm ₅ MgAlO	1.97/1.51	1.3	66.3	3.1	6.6	0.86
Yb ₅ MgAlO	1.68/1.54	1.2	53	4.1	15.1	0.64

[a] Reaction conditions: 0.5 g catalyst, 39.5 g ethanol, 473 K, autogenic pressure, reaction time 5 h.

**Scheme 3.** The formation mechanism of n-butanol on pairs of adjacent Lewis acid-basic sites.

molecule initially adsorbs on Lewis acid center (M^{+}) and adjacent basic center (O^{2-}) pairs to generate intermediate species (ethoxide), which then undergoes H removal from C_{α} -H to form acetaldehyde. Subsequently, aldol condensation of acetaldehyde occurs on adjacent Lewis acid and strong basic sites. The former adsorbs one acetaldehyde molecule and the latter is related to a proton abstraction generating carbon anion intermediate species which attacks another acetaldehyde molecule to produce 3-hydroxybutyraldehyde, then followed by dehydration and hydrogenation to form n-butanol. This Guerbet route is subsequently verified by Carvalho et al. who conducted an *in situ* Fourier transform infrared spectroscopy (FTIR) in corporation with DRIFT-MS experiments for this ethanol upgrading process.^[26] By contrast, Lewis acid sites or strong basic sites alone rarely make

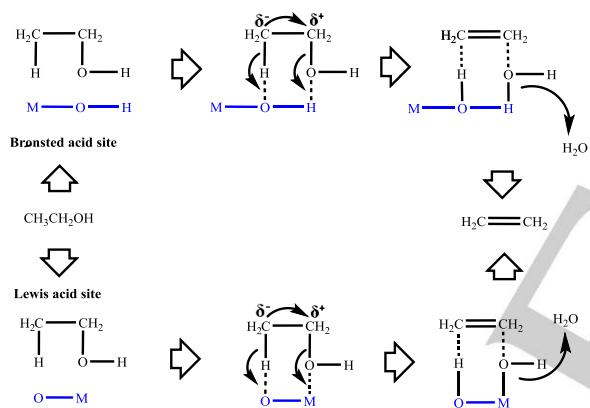
any contribution to the catalytic ethanol condensation. Actually, almost >95% ethylene selectivity was achieved over pure Al_2O_3 catalysts mainly due to the presence of a large amount of Brønsted acid ($Al-OH$) and Lewis acid centers (Al^{3+}) on the surface of Al_2O_3 catalysts.^[24] The formation mechanism of ethylene on Brønsted and Lewis acid centers is shown in Scheme 4.^[27] Although there are many strong basic sites over MgO catalysts, no n-butanol and C_4 products were generated (Table 3) mainly because of the lack of Lewis acid sites to stabilize ethoxide intermediates in the course of reaction.^[25] A distinct conclusion can be drawn that the appropriate balance in quantity between Lewis acid and basic sites plays a crucial role in improving catalytic performance, in which Lewis acid centers seem to be more important. In fact, compared to Al_2O_3 and MgO catalysts, adjacent Lewis acid and medium basic sites over Mg-Al mixed oxide catalysts are able to promote the synthesis of C_4 compounds from ethanol. In this case, strong basic sites seem not to be essential for the synthesis of C_4 products, if Lewis acid centers are present. Therefore, a further investigation on the role of Lewis acid centers is required in the future.

Based on the study on Mg-Al mixed oxide catalysts, Tanchoux et al. reported M-Mg-Al-O (M=Cu, Pd, Ag, Mn, Fe, Sm, Yb) mixed oxide catalysts for the ethanol conversion to n-butanol.^[28] These catalysts showed different catalytic activities and selectivities to n-butanol as in Table 4. In fact, the catalytic performance largely depended on acid-basic properties of these catalysts when the different metals were introduced in mixed oxide catalysts. In this context, both Cu- and Pd-based catalysts were the most active in terms of both ethanol conversion and n-butanol selectivity. For Cu-Mg-Al-O catalysts, with the increase in Cu loading, the number of basic sites decreased while the selectivity to n-butanol decreased. This result indicates the formation of n-butanol is closely related to the number of basic sites. Moreover, the decrease in the number of basic sites probably also raised the density of Brønsted acid sites on the

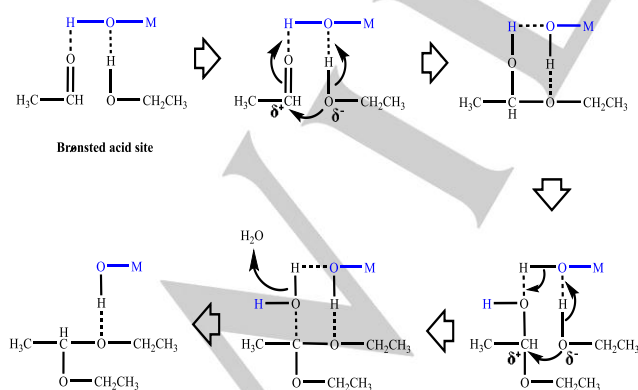
MINIREVIEW

surface of catalysts, which resulted in the production of 1,1-diethoxyethane as the main by-product. The mechanism for the formation of 1,1-diethoxyethane on Brønsted acid sites is shown in Scheme 5.^[29]

Both Pd₅MgAlO and Sm₅MgAlO catalysts showed good selectivity to n-butanol, because they had similar number of basic sites. But Sm₅MgAlO catalyst was less selective than Pd₅MgAlO, probably suggesting the synergy between basic and acid sites was better achieved in the case of Pd₅MgAlO catalyst. In addition, raising reaction temperature has a positive effect on improving the catalytic performance of Pd₅MgAlO catalyst. The ethanol conversion and n-butanol selectivity increased from 4.2% and 72.7% to 17.5% and 78%, respectively, as the reaction temperature increased from 473 to 533 K. It should be mentioned that water produced in the reaction system has a negative effect on catalytic performance probably due to the transformation of uncoordinated Lewis basic O²⁻ which is the active site for aldol condensation of acetaldehyde into weaker basic OH⁻.^[30] Therefore, an attempt should be made to remove the water from reaction system (probably by raising reaction temperature), thus enhancing the catalytic activity and further facilitating the



Scheme 4. The formation mechanism of ethylene on Brønsted and Lewis acid centers.



Scheme 5. The formation mechanism of 1,1-diethoxyethane on Brønsted acid centers.

investigation of the mechanism of acid-base catalyzed reaction.

3.2. Hydroxyapatite (HAP) catalysts

Hydroxyapatite (HAP) is generally used for catalyzing alcohol dehydrogenation, hydrogen transfer and aldol condensation of aldehyde in recent years due to its superior acid-basic bifunctional properties in a single crystal lattice.^[31] Based on the specific character of HAP, Ogo et al. reported four substituted HAP catalysts, namely Sr₁₀(PO₄)₆(OH)₂ (Sr-HAP), Ca₁₀(PO₄)₆(OH)₂ (Ca-HAP), Sr₁₀(VO₄)₆(OH)₂ (Sr-V-HAP) and Ca₁₀(VO₄)₆(OH)₂ (Ca-V-HAP), for ethanol conversion to n-butanol (573 K, atmospheric pressure, W/F=130 h·g·mol⁻¹).^[32] n-butanol was the major product over the Sr-HAP and Ca-HAP catalysts, while substantial side products (e.g. ethylene, C₄-olefins and acetaldehyde) were detected over Sr-V-HAP and Ca-V-HAP catalysts. Furthermore, the Sr-HAP catalyst exhibited up to 81.2% selectivity to n-butanol relative to 74.5% selectivity to n-butanol for Ca-HAP catalyst. This improved catalytic performance for Sr-HAP catalyst is probably attributed to its higher selectivity to crotonaldehyde in the aldol condensation of acetaldehyde and higher catalytic ability in the hydrogen transfer reaction of 2-buten-1-ol to n-butanol compared to those of other three HAP catalysts. Subsequently, to further improve catalytic performance over HAP or substituted HAP catalysts, Ueda and Ogo et al. respectively employed Ca-HAP and Sr-HAP catalysts with different Ca/P and Sr/P molar ratios for ethanol coupling to n-butanol.^[33] The catalytic performance is shown in Table 5. It can be seen that the corresponding density of basic sites increased with the increase in Ca/P or Sr/P molar ratio of HAP catalysts, followed by the increase in n-butanol selectivity. This result suggests that the n-butanol formation largely depends on the basic density. Furthermore, maximum 86.4% selectivity to n-butanol which had not been ever reported over heterogeneous catalysts was achieved at the highest Sr/P molar ratio of 1.70 and highest density of basic sites. However, this high selectivity to n-butanol is based upon high W/F_{ethanol} and reaction temperature, which means long contact time was required and a lot of energy was consumed at the same time. In addition, the catalyst tended to deactivate at higher contact time as acetaldehyde is highly reactive and can lead to cyclic compounds via self-aldolization and subsequent coke formation. As expected, the formation of ethylene is mainly attributed to higher density of Brønsted acid sites^[27] resulted from P-OH groups and/or a small amount of Lewis acid Ca²⁺ sites due to the low Ca/P molar ratio. Thus, a suitable balance between the density of acidic centers and that of basic centers is required as the case of Mg-Al mixed oxide catalysts. In addition, the influence of reaction temperature on catalytic performance has been investigated by Tsuchida et al.^[34] who employed a HAP catalyst with Ca/P molar ratio of 1.64 at a constant contact time of 1.78 s. Ethanol conversion significantly increased from 14.7 to 95.3% when the reaction temperature increased from 573 to 723 K. However, the selectivity to n-butanol rapidly decreased from 76.3 to 6.0%. This result suggests an optimal reaction temperature (≤573 K) is required to perform ethanol upgrading to n-butanol instead of production of unwanted by-products.

MINIREVIEW

Table 5. The catalytic activities and selectivities to n-butanol and by-products over HAP catalysts with different densities of acid and basic sites.^[a]

Catalyst	Density of acid sites ($\mu\text{mol}/\text{m}^2$)	Density of basic sites ($\mu\text{mol}/\text{m}^2$)	Ethanol conversion (%)	Selectivity (%)		
				Ethylene	Acetaldehyde	n-Butanol
Ca-HAP-1 (1.59)	0.038	0.01	16.2	29.2	9.4	22.2
Ca-HAP-2 (1.62)	0.029	0.02	20.8	8.6	10.2	50.4
Ca-HAP-3 (1.65)	0.011	0.38	21.2	4.7	15.4	62.4
Ca-HAP-4 (1.67)	0.006	0.53	15.8	2.8	18.3	56.2
Sr-HAP-1 (1.58)	<0.01	0.37	1.1	-	1.5	69
Sr-HAP-2 (1.64)	0.13	0.58	5.9	-	0.7	78.1
Sr-HAP-3 (1.67)	0.11	0.79	7.9	-	0.5	81.7
Sr-HAP-4 (1.70)	0.22	1	11.3	-	0.7	86.4

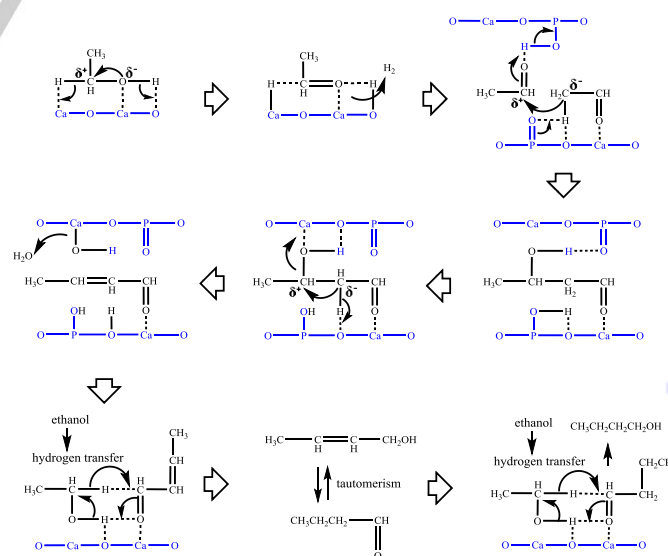
[a] Reaction conditions for the Ca-HAP series catalysts: 673 K, atmospheric pressure, GHSV: 10000 h^{-1} , 20% ethanol-80% He (vol.%); Reaction conditions for the Sr-HAP series catalysts: 573 K, atmospheric pressure, $\text{W}/\text{F}_{\text{ethanol}} = 130 \text{ h} \cdot \text{g} \cdot \text{mol}^{-1}$, 20% ethanol-80% He (vol.%). The Ca/P or Sr/P molar ratio of HAP catalysts is shown in parenthesis.

At present, there are also two types of reaction mechanism called “direct” and “indirect” route for catalyzing ethanol conversion to n-butanol over HAP catalysts.^[35] The “direct route” is related to the dimerization mechanism of two ethanol molecules, which is analogous to that shown in Scheme 2 and widely considered to be effective, ideal and environmentally friendly synthetic route of n-butanol. The prerequisite for “direct mechanism” to occur is probably the high reaction temperatures ($\geq 623 \text{ K}$), while the “indirect route” is similar to the Guerbet reaction process over acid-basic bifunctional catalysts at lower reaction temperatures. However, it is difficult to completely rule out the “indirect route” even at high reaction temperatures ($\geq 623 \text{ K}$), because these two reaction pathways probably take place simultaneously in a reaction system over hydroxyapatite catalysts.

Recently, Bell et al.^[36] proposed a detailed reaction mechanism (shown in Scheme 6) proved by *in situ* titration and spectroscopic experiments for ethanol conversion to n-butanol over HAP catalysts. There exist two kinds of different active centers: basic Ca-O sites catalyzing the dehydrogenation of ethanol and $\text{CaO}/\text{PO}_4^{3-}$ pairs responsible for the aldol condensation of acetaldehyde. This mechanism mainly includes the formation of acetaldehyde on basic Ca-O active centers and then migration to adjacent $\text{CaO}/\text{PO}_4^{3-}$ pairs to form crotonaldehyde and subsequent hydrogenation of crotonaldehyde by hydrogen transferred from ethanol on basic Ca-O sites. The key intermediates such as acetaldehyde, crotonaldehyde, crotyl alcohol and n-butanol were observed after reaction at 603 K. And the measurement of reaction kinetics at various ethanol and acetaldehyde partial pressures further suggested that a mechanism somewhat similar Guerbet pathway (Scheme 1b) could occur for ethanol coupling to n-butanol. The main difference was that the intermediate and final hydrogenation of intermediates was not achieved through H_2 , but through H-transfer from ethanol, as suggested by many authors.^[35,37] This subtle difference is of importance and suggest an interplay of mechanism in the presence of metal particles (able to activate H_2). This statement has been further verified by the fact that the hydrogenation of crotonaldehyde seldom occurs on HAP

catalysts when gaseous H_2 is added to the reaction system.^[38] The lack of Brønsted acid centers proved by FTIR spectroscopy clearly indicates Brønsted acid sites seem not to play any role in assisting the synthesis of n-butanol but to catalyze the formation of many by-products, such as ethylene and 1,1-diethoxyethane as previously mentioned above.

In conclusion, the catalytic active centers for ethanol coupling to n-butanol are still adjacent Lewis acid-base pairs over HAP catalysts. This catalytic reaction could actually follow a pathway similar to the Guerbet route, differing only in the way intermediate hydrogenation is achieved. Although the hydrogen source has been identified, the way to promote the hydrogen transfer and suppress gaseous H_2 formation from ethanol has not been suggested. More investigations on this aspect will lead to the design of more effective catalysts for crotonaldehyde hydrogenation and consequent high yield of n-butanol.

**Scheme 6.** The formation mechanism of n-butanol over HAP catalysts.

MINIREVIEW

Table 6. Catalytic performance of alumina supported different metals in the ethanol coupling process.^[a]

Catalyst	<i>d</i> band center, ϵ_d (eV)	Ethanol conversion (%)	Selectivity (%)				
			Acetaldehyde	Diethyl ether	Ethyl acetate	n-Butanol	1,1-Diethoxyethane
5%Ru/Al ₂ O ₃	-1.41	12	54	3	2	9	31
5%Rh/Al ₂ O ₃ ^[b]	-1.73	5	4	41	0	35	4
5%Pd/Al ₂ O ₃ ^[b]	-1.83	9	3	64	1	21	2
5%Pt/Al ₂ O ₃ ^[b]	-2.25	3	9	10	9	37	8
0.8%Au/Al ₂ O ₃ ^[b]	-3.56	6	18	31	15	35	0
6%Ag/Al ₂ O ₃	-4.30	2	45	11	6	20	16

[a] Reaction conditions: 0.01-0.05 g catalyst, 1.2 g ethanol, 523 K, autogenic pressure, reaction time 3 h; [b] Commercial catalysts.

3.3. Supported metal catalysts

Application of supported metal catalysts in the conversion of ethanol to n-butanol has been more and more frequently observed in recent years, due to the fact that supported metal catalysts could significantly improve the conversion of ethanol and decrease the reaction temperature, which is mainly ascribed to the enhanced ethanol dehydrogenation on metals. Thus, for this tandem catalytic reaction process different active centers are proposed for each individual reaction. Typically, metals are responsible for a primary alcohol dehydrogenation and unsaturated aldehydes hydrogenation,^[39] while adjacent Lewis acid-base pairs provide the active centers for aldol condensation of acetaldehyde and subsequent dehydration.

The ability of metal for a primary alcohol dehydrogenation depends on its ability to adsorb atomic hydrogen, which is the result of its *d* states. In general, the hydrogen adsorption energy varies with the position of the *d* band center relative to the Fermi level (ϵ_d), namely, the upshift the *d* band center can increase the reactivity of the metal and the hydrogen-metal interaction.^[40] Based on this concept, we expect there is a correlation between the *d* band center (ϵ_d) and the reactivity of metal towards alcohol dehydrogenation. Riitonen and co-authors^[41] employed a series of different self-prepared and commercial supported metal catalysts M/Al₂O₃ (M=Ru, Rh, Pd, Pt, Au, Ni) for ethanol conversion to n-butanol. As shown in Table 6, with comparable metal particle size and weight loading, a general trend that the ethanol conversion decreases as the downshift of the *d* band center of Ru to Ag metal can be seen. Although these noble metals are active in ethanol dehydrogenation reaction, a significant amount of by-product diethyl ether was generated. Additionally, the high cost and low availability of noble metals become the major barriers towards their industrial viability.

The same research group has then conducted this reaction using different non-noble metals (Cu, Co and Ni) supported on aluminum oxide in a continuous fixed-bed reactor (513 K, 70 bar, LHSV=4.3 h⁻¹).^[42] As shown in Table 7, a correlation between the *d* band center of these non-noble metals and their ability towards ethanol dehydrogenation seems still valid. Besides, they found the crystal structure of metals also played a critical role in determining the product selectivity. Catalytically active sites for n-

butanol formation are based on inverse spinel metal aluminate structure containing octahedrally coordinated metal cations, while tetrahedrally coordinated cations result in the formation of ethyl acetate. In this case, Cu and Ni catalysts exhibited better selectivity towards n-butanol than the Co catalyst which, on the contrary, displayed high selectivity towards ethyl acetate (Table 7). The difference in the composition of cations in metals was also suggested to influence the density of acid sites, which might in turn influence the adjacent Lewis acid-base pairs that supposed to be the active sites as previously discussed over the Mg-Al mixed oxide and HAP catalysts. However, the underlying mechanism for the dependence of selectivity on the metal crystal structure needs to be further revealed, and the differentiation between the effect of metal crystal structure and that of Lewis acid-base pairs on the product selectivity should be made.

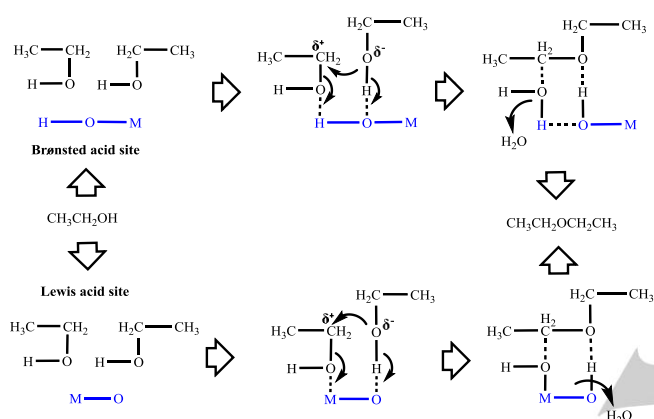
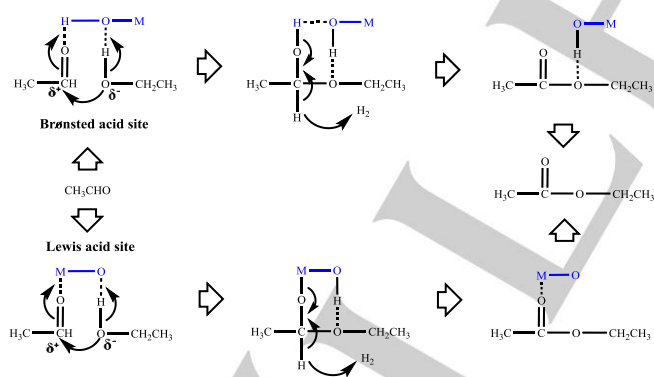
Recently, Zheng et al.^[43] reported highly dispersed Ni-MgAlO catalysts for ethanol conversion to high-carbon alcohols. The optimal catalyst Ni₄MgAlO exhibited 55.2% of selectivity to n-butanol and 85% of that to C₄-C₈ alcohols, respectively, at 18.7% ethanol conversion and reaction temperature of 523 K. The characterization results showed that metallic Ni sites promoted ethanol dehydrogenation, and the *in situ* formed hydrogen spilled over the catalyst for the subsequent crotonaldehyde hydrogenation. In this case, both ethanol conversion and n-butanol selectivity could be boosted. This finding highlights the aspect which is different over metal-free catalysts, that n-butanol selectivity can be not only affected by Lewis acid-base pairs, but also by metal's capability of hydrogen transfer and hydrogenation.

It should be noted that the n-butanol selectivity over Ni/Al₂O₃ is calculated only based on liquid carbon products. In fact, a substantial amount of gaseous products such as CO, CO₂ and CH₄ could be detected due to the strong cracking capability of the C-C bond of ethanol over Ni metal. Moreover, other unwanted by-products such as diethyl ether and ethyl acetate were also collected mainly due to the presence of Lewis and Brønsted acid sites on the surface of aluminum oxide. The mechanism for the formation of diethyl ether^[44] and ethyl acetate^[45] are shown in Scheme 7 and 8, respectively. With the attempts to reduce the formation of diethyl ether and ethyl acetate, and increase the selectivity towards the products via aldol condensation, La₂O₃ modified 8%Ni/Al₂O₃ was applied by Jordison and co-workers.^[46]

MINIREVIEW

Table 7. Catalytic performance and corresponding crystal structure of different metal catalysts.^[a]

Catalyst	d band center, ϵ_d (eV)	Ethanol Conversion (%)	Selectivity (%)		Lewis acid sites number (mmol/g)	Brønsted acid sites number (mmol/g)	Corresponding structure
			Ethyl acetate	n-Butanol			
16% Co/Al ₂ O ₃	-1.17	28	85	4	76	14	Co ²⁺ (tetrahedral), Co ³⁺ (octahedral)
19% Ni/Al ₂ O ₃	-1.29	25	5	57	63	17	Ni ²⁺ (octahedral)
4.5% Cu/Al ₂ O ₃	-2.67	15	37	55	225	25	Cu ²⁺ (octahedral)

[a] Reaction conditions: 513 K, 70 bar, LHSV=4.3 h⁻¹.**Scheme 7.** The mechanism for the formation of diethyl ether on Lewis and Brønsted acid centers.**Scheme 8.** The mechanism for the formation of ether acetate on Lewis and Brønsted acid centers.

As a result, 39% higher alcohols (C₄⁺) yield and 55% ethanol conversion were obtained in comparison to 26% yield of higher alcohols and 46% conversion of ethanol over pure Ni/Al₂O₃ catalyst, which indicates that the number of basic sites is crucial for improving catalytic performance. Based on the product

distribution over Ni catalysts, the dominant reaction route for this ethanol catalytic coupling also follows the Guerbet pathway.

Although Ni-based catalysts exhibit superior selectivity to n-butanol over Ni metal is undesirable in terms of industrial applications. Comparing to Ni-based catalysts, Cu-based catalysts show weak ability to cleave the C-C bond,^[47] while they are well known as very active heterogeneous catalysts for various reactions, such as steam reforming of methanol, CO oxidation, hydrogenation and dehydrogenation reactions.^[48] Poliakoff et al.^[49] reported different solid acid supported copper catalysts for upgrading ethanol to n-butanol in the presence of supercritical CO₂. The reaction results are shown in Table 8. High surface area (HSA) CeO₂ supported Cu catalyst exhibited excellent performance with 45% selectivity to n-butanol and up to 30% yield of n-butanol. The main side products were ethyl acetate and diethyl ether due to the presence of a large amount of Brønsted and Lewis acid sites over the different solid acid supports as previously presented. Additionally, HAS-CeO₂ supported Cu catalyst in this reaction systems showed prominent stability under such a harsh reaction condition, which is attributed to the strong Cu-CeO₂ interaction. On the contrary, other Cu catalysts easily deactivated due to the weak Cu-support interaction, which should be considered as an important problem when we try to design Cu-based catalysts for this type of reaction.^[50] Moreover, using supercritical fluid CO₂ as reaction medium^[51] had a positive effect on the maintenance of active Ce⁴⁺ species for aldol condensation in the reaction system, probably through a redox cycle in which Ce⁴⁺ was reduced to Ce³⁺ by the H₂ derived from ethanol dehydrogenation and then regenerated by the supercritical CO₂.^[52] However, the supercritical reaction pressure should be higher than 10 MPa, which would require special reaction equipment and reduce the production capacity of n-butanol in per unit volume reactor.

More recently, a series of Cu-CeO₂/AC (AC: activated carbon) catalysts with different molar ratios of Cu/Ce were used for continuous upgrading of ethanol to n-butanol in our group, wherein these catalysts exhibited nearly 20% n-butanol yield and good stability in 12 h continuous evaluation (523 K, 2 MPa, LHSV=4 ml·h⁻¹·g_{cat}⁻¹).^[53] The catalytic reaction results are shown in Table 9. The high catalytic activities and selectivities to n-butanol are mainly ascribed to the large number of highly dispersed adjacent Cu and CeO₂ centers for dehydrogenation/hydrogenation and aldol condensation over the Cu-CeO₂/AC catalysts, respectively. The good stability is because the addition of Ce species effectively restricted the migration and

MINIREVIEW

Table 8. The catalytic performance over different solid acid supported Cu catalysts.^[a]

Catalyst	Ethanol conversion (%)	Yield (%)			
		Acetaldehyde	Ethyl acetate	Diethylether	n-Butanol
Cu/Al ₂ O ₃	48	2.2	28	3.8	3
Cu/ZSM-5	15	6.5	4.1	12.4	0
Cu/CeO ₂	39	1.2	7.2	0.2	13
Cu/HSA-CeO ₂	67	0.3	3.2	0.1	30
Cu/TiO ₂	53	3.1	13.2	1.2	13
Cu/SiO ₂ -Al ₂ O ₃	30	6.1	12.8	8.2	6

[a] Reaction conditions: 533 K, 100 bar supercritical CO₂, LHSV = 1.97 h⁻¹.

growth of Cu nanoparticles on the surface of AC support mainly by partially covering Cu nanoparticles (in the case of low Ce loading) and further isolating Cu nanoparticles (in the case of high Ce loading).

It can be also seen in Table 9, the main products over the Cu-CeO₂/AC catalysts were acetaldehyde, butyraldehyde, ethyl acetate, n-butanol, hexaldehyde, 2-ethyl butanol and 1-hexanol, which typically accounted for over 90% of all the products. This kind of product distribution indicates n-butanol was produced from ethanol via a sequence of steps collectively known as the Guerbet reaction pathway. The low catalytic activity and selectivity to n-butanol over the Cu/AC catalyst should be largely ascribed to the absence of active basic sites responsible for the aldol condensation. There was a large number of active basic sites produced with the addition of Ce component and therefore the activity and selectivity to n-butanol dramatically increased over the Cu-CeO₂/AC catalysts, which further confirms basic sites are essential and active for the synthesis of n-butanol. Kinetic

analysis revealed the apparent activation energy for ethanol dehydrogenation, aldol condensation and hydrogenation is 32.5, 70.1, 12.4 kJ/mol, respectively, indicating aldol condensation is the decisive step in these tandem steps for n-butanol production.

In addition, possible gaseous products (via ethanol cracking decarbonylation of acetaldehyde) such as CH₄, CO and CO₂ were not detected in the tail-gas, further confirming the lower ability of Cu metal for C-C bond cleavage than Ni metal. Additionally, ethyl acetate, a main by-product, was dominant over the 3Cu-1CeO₂/SiO₂ or 3Cu-1CeO₂/Al₂O₃ catalyst, probably because the presence of many Lewis and Brønsted acid sites on the surface of these catalysts.^[45] An another interpretation for ethyl acetate formation is that there were more Cu⁺ species existing in the reduced Cu-CeO₂/SiO₂ and Cu-CeO₂/Al₂O₃ catalysts due to the strong Cu-support interaction, which are considered as main catalytic active centers for the formation of ethyl acetate.^[54]

Table 9. Catalytic performance of Cu-CeO₂/AC catalysts with the different Cu/Ce molar ratios.^[a]

Catalyst	Conversion (%)	Selectivity(%)						Butanol Yield (%)
		Acetaldehyde	Butyraldehyde	Ethyl Acetate	Butanol	C6 products ^[b]	Others ^[c]	
Cu/AC	1.2	10.5	0.5	0.5	12.1	-	76.4	0.2
5Cu1Ce/AC	46.2	6.8	8.2	10.3	41.2	20.6	12.9	19.0
4Cu1Ce/AC	45.6	7.8	7.8	8.5	42.4	20.3	12.8	19.3
3Cu1Ce/AC	46.2	7.2	7.8	10.9	41.3	19.7	13.0	19.1
2Cu1Ce/AC	46.3	6.6	7.5	12.5	39.3	19.4	14.6	18.2
1Cu1Ce/AC	44.0	5.6	6.7	17.4	36.8	8.7	24.8	16.3
3Cu1Ce/SiO ₂	23.3	17.1	17.0	45.4	10.2	2.1	8.1	2.4
3Cu1Ce/Al ₂ O ₃	46.9	4.4	16.7	44.7	12.1	5.3	16.8	5.7
3Cu1Ce/AC ^[d]	39.1	1.1	1.8	6.3	55.2	20.3	15.2	21.6

[a] Conversion, selectivity and yield are obtained at steady-state; reaction conditions: 1.0 g catalyst; 523 K, 2 MPa (N₂), LHSV=4 ml/(h·g_{cat}), N₂/ethanol(v/v)=500:1; [b] C6 products include 2-ethylbutyraldehyde, hexaldehyde, 2-ethylbutanol, and 1-hexanol; [c] Other products include 1-ethoxyethane, 1,1-diethoxyethane, butyl acetate, etc; specially for Cu/AC, 2.6% 1-ethoxyethane, 50.2% 1,1-diethoxyethane; for Cu-CeO₂/AC catalysts with Cu/Ce molar ratio of 1:1, 5.0% 1-ethoxyethane, 9.3% 1,1-diethoxyethane. [d] Reaction was carried out in a batch reactor with volume of 250 mL. Reaction conditions: catalyst, 3.3 g (<120 mesh; 80 g absolute ethanol; 523 K, 0.1 MPa (N₂); reaction time 48 h.

MINIREVIEW

Table 10. Comparison of catalytic performance of Cu-CeO₂/AC catalysts with other catalysts reported in documentary.

Catalyst	T (K)	P (MPa)	Other reaction conditions	Ethanol Conversion (%)	Butanol Selectivity (%)	Butanol Yield (%)	Conversion · liquid hourly space velocity (10 ⁻² mol·g ⁻¹ ·h ⁻¹)	Ref.
Rb-NaX zeolite	693	0.1	W/F=5.6 g·h·mol ⁻¹	0.21	40.9	0.86	0.04	[19]
MgO	723	0.1	0.5 g catalyst W/F=20.83 g·h·mol ⁻¹	56.14	32.76	18.39	2.69	[17]
3Mg1AlO	623	0.1	W/F=23.33 g·h·mol ⁻¹	33.2	34.3	11.39	1.42	[22]
HAP (Ca/P=1.64)	573	0.1	Contact time = 1.78 s	14.7	76.3	11.22	Unknown	[21]
HAP (Sr/P=1.70)	573	0.1	W/F=130 g·h·mol ⁻¹	11.3	86.4	9.76	0.09	[23]
HTC-500 20.7%Ni/Al ₂ O ₃	513	7	W/F=11.11 g·h·mol ⁻¹	10	69	6.9	0.90	[24]
Cu/HSACeO ₂	533	10	LHSV=1.97 h ⁻¹	67	44.8	30	-	[25]
Cu-CeO ₂ /AC	523	2	W/F=14.30 g·h·mol ⁻¹ (LHSV=2 h ⁻¹)	45.6	42.4	19.3	3.18	[26]
Pd-Mg-Al-O	473	0.1	0.5 g catalyst 39.5 g ethanol Reaction time 5 h	3.8	72.7	2.76	1.31	[27]
HTC-500 20.7%Ni/Al ₂ O ₃	523	7	3.3 g catalyst 80 g ethanol reaction time =72 h	25	80	20	0.18	[28]
Cu-CeO ₂ /AC	523	0.1	3.3 g catalyst 80 g ethanol reaction time = 12 h	35.1	49.8	17.5	1.54	[29]
Cu-CeO ₂ /AC	523	0.1	3.3 g catalyst 80 g ethanol reaction time = 48 h	39.1	55.2	21.6	0.43	[30]

A systematic comparison of catalytic performance associated with heterogeneous catalysts in the ethanol upgrading process has been presented in Table 10. It can be seen that the Cu-CeO₂/AC catalyst exhibited the highest reaction rate and n-butanol yield among all the catalysts in fixed-bed reactors. Furthermore, the Cu-CeO₂/AC catalyst also exhibited the highest reaction rate in batch reactors after 12 h of reaction, and the highest n-butanol yield of 21.6% was obtained after 48 h of reaction. As provided by kinetic analysis, the enhanced reaction rate and n-butanol yield are ascribed to the much lower apparent activation energies for ethanol dehydrogenation, aldol condensation and hydrogenation. The excellent catalytic performance of Cu-CeO₂/AC catalysts under such milder reaction conditions provided a promising prospect for their commercial applications.

4. Research prospects

As the most widely used catalysts in industry, heterogeneous catalysts have been extensively investigated for n-butanol synthesis from ethanol in the last decades. However, present studies conducted with metal oxides^[19,20,23-25,28,43], zeolites^[21] and hydroxyapatite^[32-36] catalysts mainly focused on the evaluation of

bulk materials under relatively harsh conditions. There are limited investigations available on the exploitation of nanoscale catalysts in ethanol upgrading to n-butanol. Comparing to bulk materials, nanostructured and nanoscale catalysts provide an important foundation for the study of structure-performance relationship, which is critical for better understanding the reaction mechanism in ethanol upgrading process. In particular, application of highly dispersed nanoparticles on high surface area materials in this reaction makes it possible to achieve optimal catalytic performances for future commercial applications. In our recent work,^[53] the high surface area AC supported Cu (Cu < 10 nm) and CeO₂ nanoparticle (< 5 nm) catalysts showed over 40% selectivity to n-butanol and nearly 20% yield of n-butanol under mild reaction conditions: 523 K, 2 MPa (N₂), LHSV = 4 ml/(h·g_{cat}) and N₂/ethanol (v/v) = 500:1. This improved catalytic performance is mainly attributed to the high dispersion of Cu and CeO₂, and the synergy of Cu, CeO₂ and activated carbon support. Additionally, the addition of Ce species stabilized Cu nanoparticles on the surface of activated carbon by partially covering Cu nanoparticles (in the case of low Ce loading) and further isolating Cu nanoparticles (in the case of high Ce loading). Further investigations are suggested to focus on the design and preparation of nano-catalysts with Cu and CeO₂ having specific structures (such as various particle sizes, preferentially exposed

MINIREVIEW

facets, morphology, spatial position, etc.), so that the study of structure-performance relationship can be easily performed, and based on which more efficient and stable catalysts for ethanol upgrading to n-butanol can be synthesized.

In the Guerbert reaction chain, there are four independent reaction processes including ethanol dehydrogenation, aldol condensation of acetaldehyde, dehydration of 3-hydroxybutyraldehyde and hydrogenation of crotonaldehyde, which form a tandem catalytic reaction pathway. To boost the tandem catalytic process for upgrading ethanol, we should first identify a component that can selectively dehydrogenate ethanol, preferably accompanied with strong ability of hydrogen transfer, and simultaneously is active in the hydrogenation of unsaturated aldehydes. The active centers for dehydrogenation are generally transition metals such as Cu, Ni, Pt, Pd, etc., among which non-noble metal Cu and Ni may be more suitable in terms of future industrial applications. However, Cu and Ni have relative low catalytic activity for hydrogenation. Our recent work demonstrated that the Pd-CeO₂/AC catalyst had higher selectivity towards n-butanol (67.6%) but lower ethanol dehydrogenation activity (11.9%) than the Cu-CeO₂/AC and Ni-CeO₂/AC catalysts.^[55] Thus, the combination of two metals in the same catalyst, one being active in ethanol dehydrogenation (such as Cu and Ni) and the other in hydrogenation steps (such as Pd), might be considered as a promising approach to designing efficient catalysts for the production of n-butanol from ethanol.

Based on the previous study, the aldol condensation of acetaldehyde is considered as the rate-determining step, while the adjacent Lewis acid-base pairs are responsible for this key step. In view of this, an interest in exploring rare earth metal oxides (La₂O₃ and CeO₂) as basic modifier to catalysts for ethanol conversion to n-butanol has increased considerably in recent years.^[30,46,49,53,55,56] However, the attention should be also paid to the development of suitable Lewis acid sites, as the requirement for the presence of strong basic sites can be alleviated, if Lewis acid centers are present. Moreover, the Lewis acid-base pairs, if introduced by the modifier or support in the catalysts, would be affected by the properties of metal loaded. For instance, the metal with high ability of hydrogen activation and spillover would facilitate the reduction of reducible oxides in the vicinity, thus changing the Lewis acidity of the catalysts.^[57] The loaded metal itself would also exhibit some extent of Lewis acidity when interacting with the support.^[58] Therefore, more investigations on the approach to finely tune the properties of Lewis acid-base pairs and their roles in the ethanol upgrading to n-butanol should be further carried out in future.

Another aspect of the catalysts for the title reaction is their selectivity and stability. As mentioned above, although Ni metal is active for ethanol dehydrogenation, it is followed by the formation of considerable amount of gaseous carbon products (via ethanol cracking, decarbonylation of acetaldehyde, etc.) such as CO, CO₂ and CH₄, due to the strong cracking capability of the C-C bond of ethanol over Ni metal.^[41-43,46,55,56b] Concerning this point, Cu-based catalysts are more preferable in the ethanol upgrading to n-butanol because the cleavage of C-C bonds is not favorable over Cu metal.^[49,53,55] However, the greatest problem for supported Cu catalysts is the rapid sintering and growth of Cu

nanoparticles.^[59] As we know, metal particle growth is a major cause of deactivation for supported catalysts.^[60] By tuning the properties of supported metal nanoparticles, including size, composition, metal-support interaction and spatial distribution, the growth rate of nanoparticles can be lowered.^[59,61] For example, the high stability of Cu metals could be achieved by supporting them on a high surface area CeO₂ (HSA-CeO₂), which is mainly attributed to the strong metal-support interaction.^[49] In our previous work, the Cu-CeO₂/AC catalysts also showed high stability in the ethanol catalytic upgrading. This good stability is ascribed to the improved Cu-CeO₂ interaction via the formation of special Cu-CeO₂ structure.^[53] Recently, Prieto et al.^[59] found the significance of the 3D spatial location of metal nanoparticles over catalysts, in which an ideal distribution for a growth-resistant catalyst is of equally sized (minimum driving force for ripening) and maximally spaced metal nanoparticles. Thus, a remarkably enhanced stability was achieved over the Cu-Zn/SiO₂ catalyst with uniformly spatially distributed Cu metals in the methanol synthesis experiments. In addition, there are also other successfully employed approaches to stabilizing supported metal nanoparticles on catalysts, including introducing different inorganic oxides via chemical vapor deposition, atomic layer deposition, encapsulating with dendrimer, post-grafting, the formation of special core-shell and semi-embedded structures.^[62] However, effective methods of maintaining Cu nanoparticle size and understanding the particle growth mechanism under realistic reaction conditions is still a key challenge to be addressed in the future. On the other hand, catalyst coking is also a significant factor resulting in catalyst deactivation in ethanol conversion to n-butanol process. In particular, the coke deposition could be mainly generated from the ethanol dehydration/dehydrogenation and then polymerization at high reaction temperature and/or under strong acid conditions.^[32] The possibility of mitigating coke deposition and limiting side reaction interferences should be adequately explored based on the understanding of structure-performance relationship of catalysts and reaction mechanism.

The selectivity and stability of catalysts, both homogeneous and heterogeneous, can be also influenced by their ability for hydrogen transfer. Hydrogen transfer may occur by one of two mechanisms,^[63] which include surface-mediated hydrogenation or direct intermolecular hydride transfer, also known as the Meerwein-Ponndorf-Verley (MPV) reaction. In the ethanol upgrading with ethanol as the sole source of hydrogen, surface-mediated hydrogenation which is common on metals, involves activation and transfer of H atoms either from gaseous H₂ derived from ethanol or ethanol itself.^[64] The product selectivity is highly dependent on the nature of metals for the preferential hydrogenation of C=C or C=O bonds. The MPV reaction is usually catalyzed by Lewis acids or bases, with the Lewis acid-base pairs performing the best.^[63c,65] In this synergy of Lewis acid-base pairs, Lewis basic sites assist the formation of ethoxide moieties on the Lewis acid sites by abstraction of a proton from ethanol, thus facilitating the formation of a six-membered intermediate between alcohol, aldehyde, and hereby transferring β -H to complete the direct intermolecular hydride transfer.^[66] The MPV reaction is a highly chemoselective reduction method for carbonyl compounds, as a result of which C=O bonds are more preferentially

MINIREVIEW

hydrogenated than C=C bonds in aldehydes. On the other hand, hydrogenation of C=C bonds proceeds by transfer of two H atoms formed on the catalyst surface via surface-mediated hydrogenation.^[64] In this case, crotonaldehyde is rapidly hydrogenated to crotyl alcohol and ultimately n-butanol. The fast hydrogen transfer reduces unwanted side reactions such as dehydration (in the case of crotyl alcohol to butadiene) or further aldol condensation (in the case of aldehyde, crotonaldehyde and butanal to C₆ or C₈ aldehydes).^[36] If the MPV reaction is too slow, the C₆ or C₈ aldehydes may further condense, which lowers the selectivity to n-butanol and causes catalyst deactivation.^[65b] However, detailed information on the promotional effects of hydrogen transfer on the selectivity and stability of catalysts are still lacking. More research work on the use of hydrogen transfer to improve the performance of catalysts is needed.

Finally, most of the heterogeneous or homogeneous reactions are carried out in organic solvent, pure reactant or diluted gas phase rather than aqueous phase, since the presence of water containing in the initial reactant mixture or from the dehydration process is usually detrimental to catalysts.^[67] On the other hand, the use of water as a reaction medium has many advantages from economic, environmental and safety standpoints, and thus is greatly anticipated in future. Moreover, the bio-ethanol derived from mature fermentation process in water is preferable to be used directly as reactant for ethanol upgrading to n-butanol reaction. Thus, catalyst systems with strong water tolerance or hydrothermal stability are intrinsically required when considering the presence of water in the reaction process. Moreover, the effect of water as a reaction medium on product distribution and even reaction mechanism should be further discussed. In previous reports, the stability of Cu-Mg-Al mixed oxides catalysts derived from hydrotalcite precursors^[24] and Ni/La₂O₃/γ-Al₂O₃^[30] catalysts in ethanol-water upgrading was discussed. These catalysts experienced rapid deactivation with the addition of water in the pure alcohol feedstock. In contrast, the good stability of Cu-CeO₂/AC catalysts might suggest their good water tolerance or hydrothermal stability, since water was one of the primary by-products in the ethanol upgrading to n-butanol.^[53] In view that carbon materials are naturally water tolerant, while CeO₂ also exhibits strong tolerance to water,^[68] the good stability of Cu-CeO₂/AC catalysts in ethanol upgrading to n-butanol is rational.

5. Conclusion

Ethanol catalytic upgrading to n-butanol is increasingly attracting scientific and industrial interest due to numerous advantages of n-butanol over ethanol as renewable fuel additive. The discontinuity of reaction process and potential environmental pollution concerning the use of soluble bases and organic solvents to some extent limit the large-scale application of homogeneous catalysts in industry, although a high selectivity to n-butanol (> 90%) is achieved with ruthenium or iridium complexes catalysts. In this context, Mg-Al mixed oxides derived from hydrotalcite, hydroxyapatite (HAP), and especially supported metal catalysts have been preferably explored as catalytic

materials for this ethanol upgrading to n-butanol reaction. Moreover, the design and development of nanostructured and nanoscale catalysts that have superior activity and selectivity for ethanol upgrading under mild conditions is highly desired for future industrial n-butanol production. Thus, further investigations on the structure-performance relationship of heterogeneous nanostructured and nanoscale catalysts and corresponding reaction mechanism for ethanol upgrading to n-butanol are very necessary. In this case, some fundamental issues such as optimal equilibrium in the quantity of Lewis acid and basic sites, fine-modulation of metal sites for enhancing the selectivity to n-butanol instead of side products such as diethyl ether and ethyl acetate, the benefit of hydrogen transfer and the influence of water on the structure and performance of catalysts and catalytic mechanism, must be critically addressed. Another critical problem is related to the catalyst durability in the process of reaction. In this case, further explorations of the nature of the synergy in Cu, CeO₂ and carbon materials, and the maintenance of Cu nanoparticle size under realistic reaction conditions show an importance for achieving more efficient and stable catalysts for bio-ethanol valorization to n-butanol and ultimately promoting the development of ethanol economy.

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Keywords: bio-ethanol • bio-butanol • catalyst development • structure-performance relationship • reaction mechanism

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MINIREVIEW

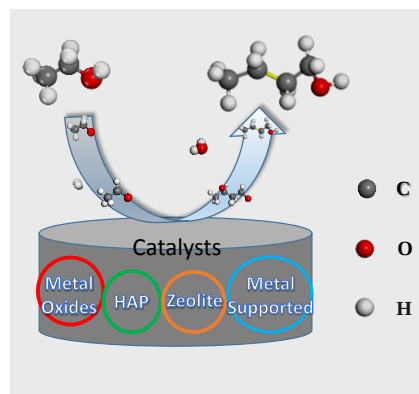
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MINIREVIEW

Table of Contents

MINIREVIEW

This paper reviewed the recent progress in catalyst development for upgrading ethanol to n-butanol, involving homogeneous and heterogeneous catalysts, with the special emphasis on the structure-performance relationship of catalysts and corresponding reaction mechanism.



Xianyuan Wu, Geqian Fang, Yuqin Tong, Dahao Jiang*, Zhe Liang, Wenhua Leng, Liu Liu, Pengxiang Tu, Hongjing Wang, Jun Ni* and Xiaonian Li

Catalytic upgrading of ethanol to n-butanol: A progress in catalyst development